

Reasoning LLMs

Module 1 — What are they, why now?

Lucas Soares · O'Reilly Live Training

What is "reasoning"?

Multi-step problem solving — not retrieval.

Vanilla LLM weak spots

- Knowledge cutoffs
- No actions in the world
- Long-horizon planning collapses

Direct answer vs. thinking chain

gpt-5.5 (low effort) — direct answer

Q: A train leaves Lisbon at 3pm going 80 km/h. Another leaves Porto at 4pm going 100 km/h. When do they meet?

A: They meet at 5:30pm.

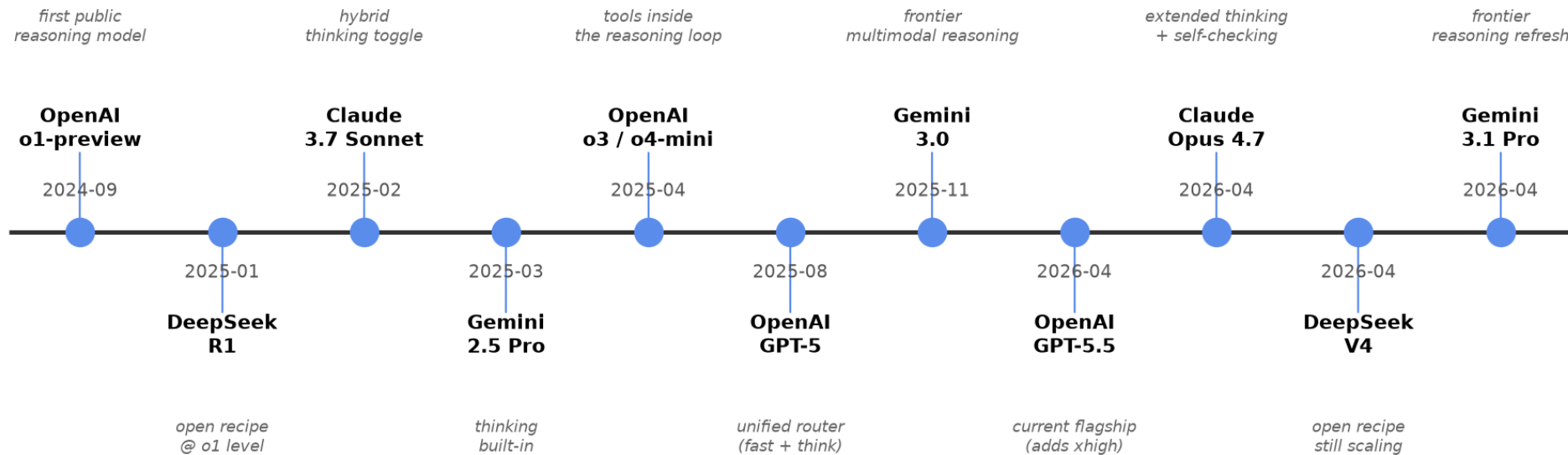
gpt-5.5 (high effort) / R1 — with <think> trace

```
<think>
Distance Lisbon↔Porto ≈ 300 km.
At 3pm train A starts. By 4pm
it has gone 80 km, so 220 km
remain. Closing speed 180 km/h.
 $t = 220/180 \approx 1.22 \text{ h} \approx 1\text{h}13\text{m}$ .
4pm + 1h13m = 5:13pm.
</think>
```

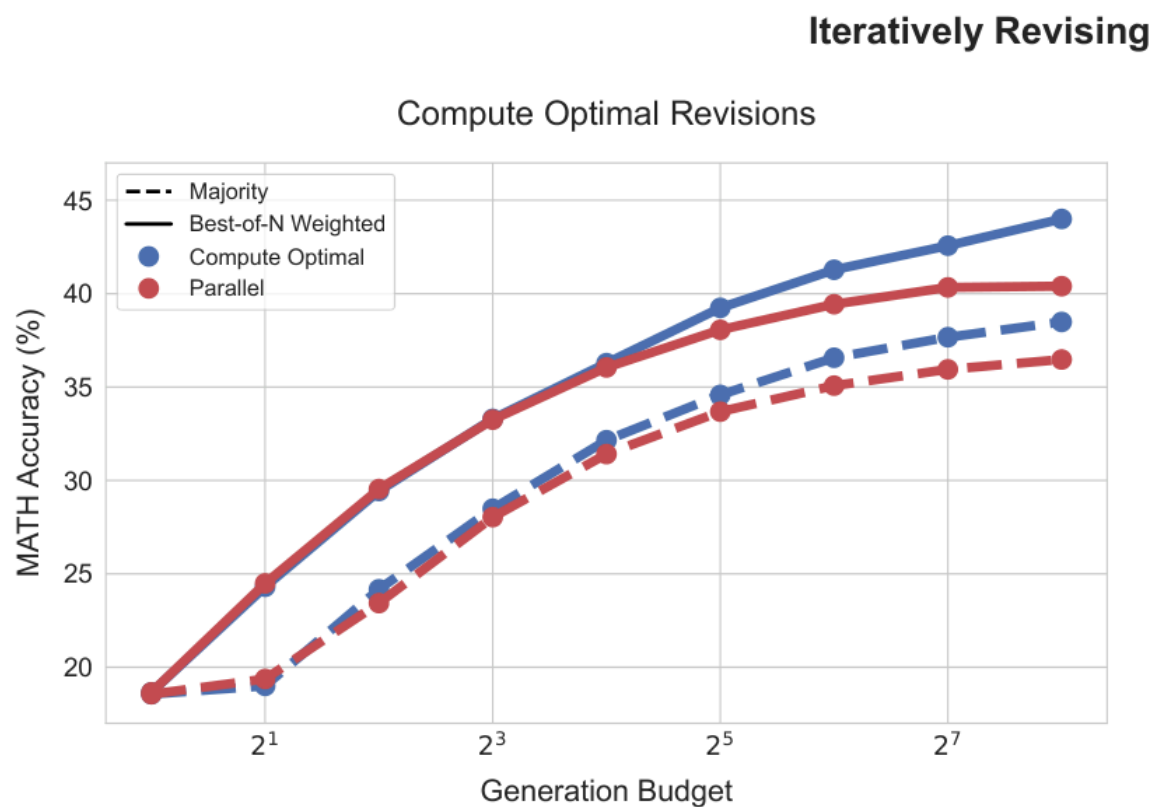
A: They meet around 5:13pm.

Timeline

The reasoning-model wave · Sep 2024 → Apr 2026



Test-time compute scaling



More thinking → more accuracy.

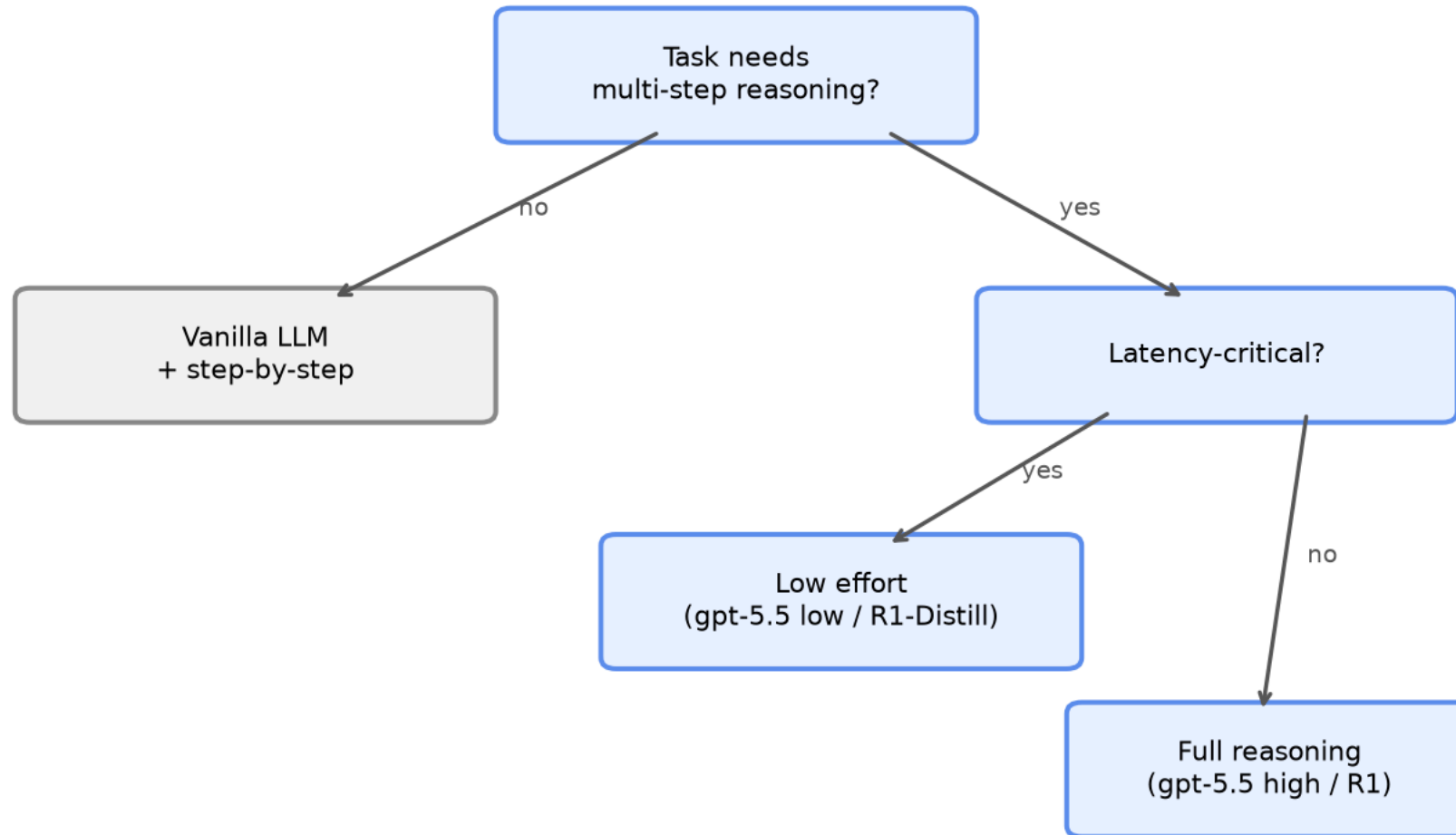
The simplest trick

Q: A train leaves at 3pm...

A: Let's think step by step.

When to reach for a reasoning model

When to reach for a reasoning model



Cost & latency

5–50× more tokens. Use only when the task needs it.

Reasoning effort knobs

- **OpenAI:** `reasoning_effort = none / low / medium / high / xhigh`
- **Anthropic:** `thinking.budget_tokens = N`

Live demo →

`notebooks/01_chain_of_thought_intuition.ipynb`

GPT-2 with and without "Let's think step by step"

Recap

- Reasoning $\neq$ retrieval
- Trade tokens for accuracy
- Next: how DeepSeek **trained** a reasoning model